

Package ‘irtempirical’

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irt_empirical_model	<i>Compute an empirical (nonparametric) IRT model</i>
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Description

Estimates item (and, for polytomous items, response-option) characteristic curves directly from raw response data, without assuming a parametric IRT form (1PL/2PL/3PL, GRM, ...). Respondents are grouped by their observed total (sum) score, which stands in for the latent trait, and for each item option the probability of endorsement is simply the empirical proportion of respondents in that score group who endorsed it. A quasibinomial logistic curve can optionally be fit through those empirical points as a smoothed approximation, and a monotonicity index is computed per option as a quick diagnostic of how well its endorsement probability tracks the trait axis.

For each item, every response option i ($1 \dots \max(\text{item})$) gets its own [EmpiricalIRT](#) entry holding three empirical curves computed from three indicator vectors: exactly option i (responses1), more than i (responses2), and less than i (responses3). For dichotomous (0/1) items there is a single option ($i = 1$) and only responses1/y1 is meaningful.

Usage

```
irt_empirical_model(data = NULL, items = NULL, addlogit = TRUE)
```

Arguments

<code>data</code>	A <code>data.frame</code> of item responses: one column per item, one row per respondent. Values must be numeric (dichotomous 0/1 or polytomous 1..k). Used to compute total scores and, per item, the empirical response-option curves.
<code>items</code>	Character vector of column names in <code>data</code> to include in the model. If <code>NULL</code> (the default), every column of <code>data</code> is used.
<code>addlogit</code>	Logical. If <code>TRUE</code> (the default), fit a quasibinomial GLM ($y \sim x$, <code>family = quasibinomial</code>) through each of the three empirical curves (y_1, y_2, y_3) against the z-score axis, so that a smoothed curve can be overlaid on the raw empirical points when plotting. If <code>FALSE</code> , curve fitting is skipped and the resulting <code>logit1/logit2/logit3</code> slots stay <code>NULL</code> .

Value

An S4 [EmpiricalIRTModel](#) object with slots:

`data` the (possibly subset-by-items) input data

`sum_scores` total score per respondent (row sum of `data`)

`sum_scores_mean, sum_scores_sd` mean and SD of `sum_scores`

`scores_axis` every integer total score from the observed minimum to maximum, i.e. the x-axis in raw score units

`z_scores_axis` `scores_axis` standardized to z-scores using `sum_scores_mean/sum_scores_sd`; this is the x-axis used for plotting and for the logistic fits

`items` names of the items included in the model

`irts` a named list, keyed by item name, of lists of [EmpiricalIRT](#) objects (one per response option of that item) — this holds all of the actual curve data described above

Use `plot(model, item = ..., ...)` to visualize the item/option characteristic curves of the returned model.

Examples

```
sim_dichotomous <- data.frame(psych::sim.irt(nvar = 10, n = 1000, low = -4, high = 4, a = NULL, c = 0, z = 1, d = NULL),
sim_polytomous <- data.frame(psych::sim.poly(nvar = 10, n = 1000, low = -4, high = 4, a = NULL, c = 0, z = 1, d = NULL),
m1 <- irt_empirical_model(data = sim_dichotomous, items = NULL)
m2 <- irt_empirical_model(data = sim_polytomous, items = NULL)
plot(m1, type = "ICC", xlim = c(-6, 6), item = "V2", draw.logit_method = 0, add_score_axis = TRUE)
plot(m2, type = "ICC", xlim = c(-6, 6), item = "V2", draw.logit_method = 0, add_score_axis = TRUE)
```

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*** assumptions**

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